THERMODYNAMICS

Course Code : 23A03301 | Handwritten Study Notes

UNIT – II : First Law of Thermodynamics

★ Note: Syllabus: Energy in state and in transition, work and heat, point and path function, zeroth law, PMM-I, Joule's experiment, first law and applications, limitations of first law, enthalpy, thermal reservoir, heat engine, heat pump, parameters of performance

1. Energy in State and Energy in Transition

Energy is the capacity to do work. It exists in two ways:

- Stored energy (energy in state) – a property of the system: kinetic energy $KE = \frac{1}{2}mV^2$, potential energy $PE = mgz$, and internal energy U.
- Energy in transition – crosses the system boundary only while a process occurs: heat (Q) and work (W). They are not stored in the system.

$$\text{Total energy } E = U + KE + PE = U + \tfrac{1}{2} m V^2 + m g z$$

Internal energy (U)

- Energy of molecules: translational, rotational, vibrational kinetic energy + intermolecular potential energy + chemical & nuclear energy.
- Extensive property, unit kJ. Specific internal energy $u = U/m$ (kJ/kg).
- For an ideal gas U depends only on temperature: $u = f(T)$, $du = c_v\, dT$.

2. Work

Work is energy transfer across the boundary due to a force acting through a distance – or, thermodynamically, any interaction other than temperature difference that can be fully converted to lifting a weight.

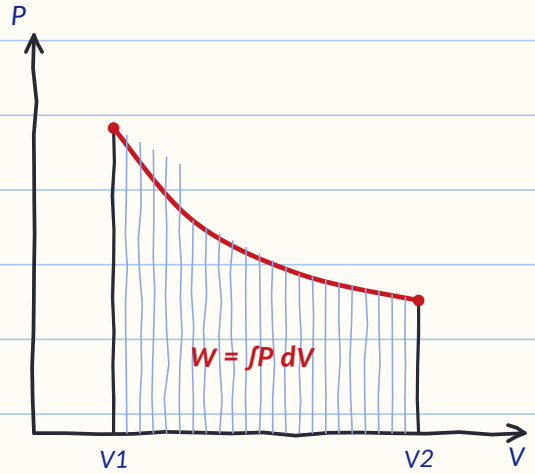
- Sign convention: work done BY the system = positive (+W); work done

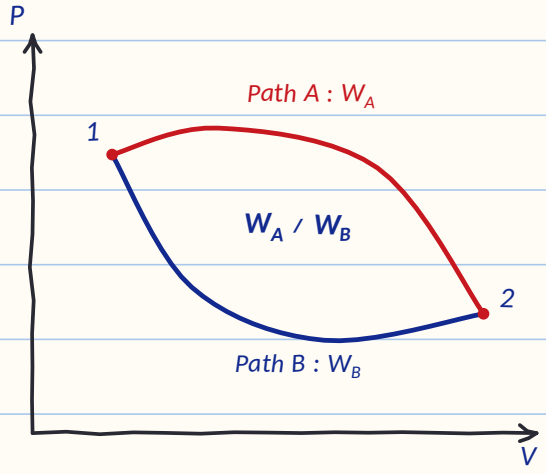
ON the system = negative (–W).

- Unit: J, kJ. Power = work per unit time (W).

Displacement (PdV) work

$$\delta W = P\,dV \qquad W_{1\text{-}2} = \int P\,dV \ \text{(from } V_1 \text{ to } V_2\text{)}$$





- Work = area under the P-V curve, valid only for quasi-static processes.

  Area depends on the path ⇒ work is a path function.

- Expansion: dV > 0, W positive. Compression: dV < 0, W negative.

Other modes of work

Mode	Expression
Shaft work	$W = T\,\omega$ (torque × angular velocity)
Electrical work	$W = V\,I\,t$
Spring work	$W = \frac{1}{2}\,k\,(x_2^2 - x_1^2)$
Stretching of wire	$W = \int \sigma\,A\,dx$
Surface film	$W = -\int \sigma\,dA$

3. Heat

Heat is energy in transit across the boundary because of a temperature difference. It flows from high to low temperature.

- Modes of heat transfer: conduction, convection, radiation.
- Sign: heat added TO the system = positive (+Q); heat rejected BY the system = negative (−Q).
- Adiabatic process: Q = 0. Heat transfer rate $\dot{Q}$ in W; specific heat transfer q = Q/m.

Heat vs Work – Comparison

Heat	Work
Due to temperature difference	Due to force × displacement
Disorganised molecular energy	Organised energy
Low-grade energy	High-grade energy
Cannot be fully converted to work	Fully convertible to heat
Both are path functions, boundary phenomena	Both are transient, not stored

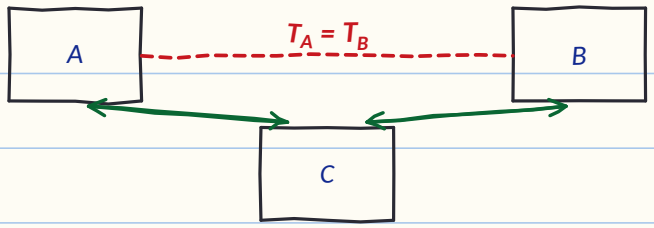
4. Point Function and Path Function

- Point function: value depends only on the end states – properties (P, T, V, U, H, S). Exact differential; $\oint dX = 0$.
- Path function: value depends on the path between end states – heat and work. Inexact differential δ; $\oint \delta Q \neq 0$, $\oint \delta W \neq 0$.

$$\int dU = U_2 - U_1 \quad \text{but} \quad \int \delta W = W_{1\text{-}2} \ (\text{NOT } W_2 - W_1)$$

5. Zeroth Law of Thermodynamics

★ Note: If two bodies are each in thermal equilibrium with a third body, then they are in thermal equilibrium with each other. (R.H. Fowler, 1931)



$T_A = T_C$ and $T_B = T_C$

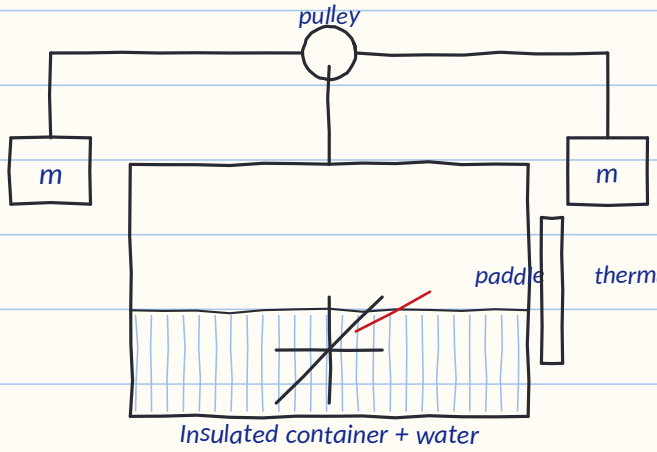
$\Rightarrow T_A = T_B$

(basis of thermometry)

- Basis for the concept of temperature – bodies in thermal equilibrium have the same temperature.
- Makes thermometers possible: thermometer (body C) reading equal for A and B $\Rightarrow T_A = T_B$.
- Called "zeroth" because it was recognised after the first and second laws but is logically more fundamental.

6. Joule's Experiment

Joule (1843-1850) converted known amounts of work into heat in an insulated vessel and measured the temperature rise of the water.



$W = 2\,m\,g\,h$

→ raises T of water

1 cal = 4.186 J

$W \propto Q$ ($J = 4.186$)

→ Falling weights turn a paddle wheel which stirs the water; work done = $m\,g\,h$ (shaft work).

→ Water temperature rises; Q-equivalent = $m_w\,c_w\,\Delta T$. The container is insulated so no heat leaves.

→ Repeated with different liquids, paddle materials, masses – the ratio W/Q remained constant.

$W \propto Q \Rightarrow W = J\,Q$, $J = 4.186$ J/cal (mechanical equivalent of heat)

- Conclusion: heat and work are mutually convertible forms of energy; in cyclic process $\oint \delta Q = \oint \delta W$ (in consistent units).

7. First Law of Thermodynamics

★ Note: Statement 1 (cycle): For a closed system undergoing a cyclic process, the net heat transfer equals the net work transfer: $\oint \delta Q = \oint \delta W$.

★ Note: Statement 2 (Clausius/Joule): Energy can neither be created nor destroyed, it can only be transformed from one form to another.

First law for a process (non-cyclic)

If a system goes from state 1 to 2 by path A and returns by path B: $\oint \delta Q = \oint \delta W \Rightarrow (Q_A - W_A) + (Q_B - W_B) = 0$. Different return paths give the same $Q-W$ $\Rightarrow Q-W$ is a property $= \Delta E$.

$$Q_{1\text{-}2} = E_2 - E_1 + W_{1\text{-}2}$$

$$\delta Q = dE + \delta W \qquad \text{(closed system)}$$

$$Q = \Delta U + W \qquad \text{(KE and PE changes neglected)}$$

- Isolated system: $Q = 0$, $W = 0 \Rightarrow \Delta E = 0$, energy of the universe is constant.
- $Q - W = \Delta E$ is also valid for irreversible processes; only the $P\,dV$ form needs quasi-static processes.

Perpetual Motion Machine of the First Kind (PMM-I)

A hypothetical machine that produces work continuously without any energy input (or delivers more energy than it receives). It violates the first law, hence impossible.

- Corollary: there is no PMM-I, $W_{cycle} = Q_{cycle}$ always.

Specific heats and ideal gas relations

$$c_v = (\partial u/\partial T)_v \qquad c_p = (\partial h/\partial T)_p$$

$$c_p - c_v = R \qquad \gamma = c_p/c_v$$

$$c_v = R/(\gamma-1) \qquad c_p = \gamma R/(\gamma-1)$$

- Air: $c_p = 1.005$, $c_v = 0.718$ kJ/kg·K, $R = 0.287$ kJ/kg·K, $\gamma = 1.4$.

8. Enthalpy

$$H = U + P\,V \qquad h = u + P\,v \quad (kJ/kg)$$

- Enthalpy is a property (U, P, V all are properties). Extensive; unit kJ.
- Physical meaning: total energy of a flowing fluid = internal energy + flow work (Pv).
- For constant-pressure quasi-static process in closed system: $Q = \Delta H$.
- Ideal gas: $h = f(T)$ only, $dh = c_p\,dT$.

9. First Law Applied to Closed Systems (Non-flow)

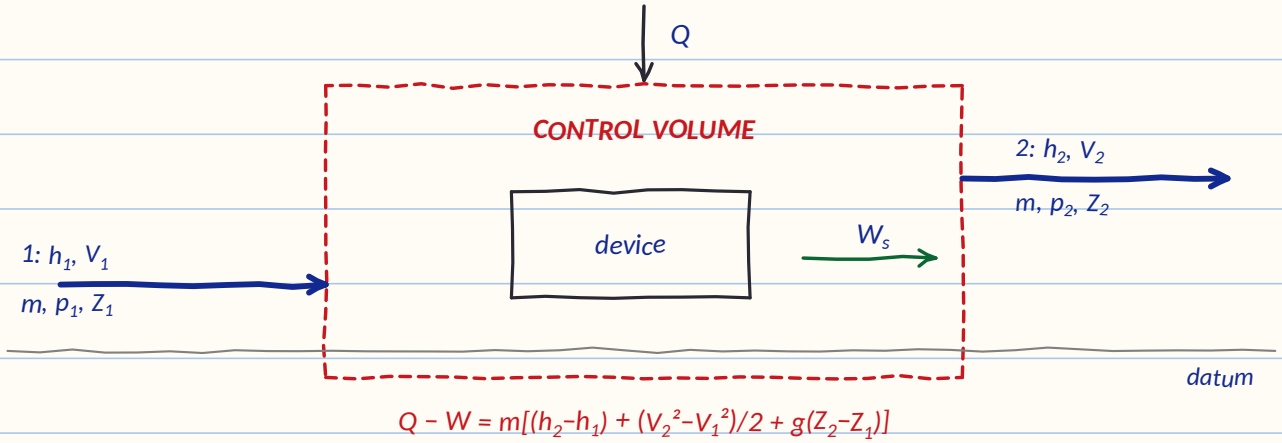
Process	W	ΔU	Q
Isochoric (V const)	0	$m\,c_v\Delta T$	ΔU
Isobaric (P const)	$P(V2-V1)=mR\Delta T$	$m\,c_v\Delta T$	$m\,c_p\Delta T = \Delta H$
Isothermal	$mRT\,\ln(V2/V1)$	0	W
Adiabatic (Q=0)	$(P1V1-P2V2)/(\gamma-1)$	−W	0
Polytropic PV^n=C	$(P1V1-P2V2)/(n-1)$	$m\,c_v\Delta T$	$W(\gamma-n)/(\gamma-1)$

$$T_2/T_1 = (V_1/V_2)^{\gamma-1} = (P_2/P_1)^{(\gamma-1)/\gamma} \quad \text{(adiabatic)}$$

- Free expansion: $Q = 0$, $W = 0 \Rightarrow \Delta U = 0$ (for ideal gas T unchanged).

10. First Law for Open Systems – SFEE

Steady Flow Energy Equation (SFEE) – steady state: mass flow in = mass flow out, properties at any point do not change with time, no accumulation of energy in the CV.



$$Q - W = m[(h_2 - h_1) + (V_2^2 - V_1^2)/2 + g(Z_2 - Z_1)]$$

$$\dot{m}_1 = \dot{m}_2 \rightarrow \rho_1 A_1 V_1 = \rho_2 A_2 V_2 \text{ (continuity)}$$

$$\dot{Q} - \dot{W}_s = \dot{m}\,[\,(h_2 - h_1) + (V_2^2 - V_1^2)/2 + g\,(z_2 - z_1)\,]$$

- Per kg: $q - w = \Delta h + \Delta KE + \Delta PE$. Flow work = P v (included in enthalpy).

Applications of SFEE

Device	Assumptions	Result
Nozzle	q=0, w=0, Δz=0	$V2 = \sqrt{[2(h1-h2)+V1^2]}$
Diffuser	q=0, w=0	V2 reduces, h2 > h1
Turbine	q≈0, ΔKE,ΔPE≈0	w = h1 – h2
Compressor/Pump	q≈0	w = h1 – h2 (negative)
Throttle valve	q=0,w=0, ΔKE≈0	h1 = h2 (isenthalpic)
Boiler	w=0	q = h2 – h1
Condenser	w=0	q = h1 – h2 rejected
Heat exchanger	overall q=0	$\dot{m}_h \Delta h_h = \dot{m}_c \Delta h_c$

11. Limitations of the First Law

→ Does not tell the direction of a process – a cup of hot coffee cooling satisfies the law, but the reverse (coffee heating itself) does not occur.

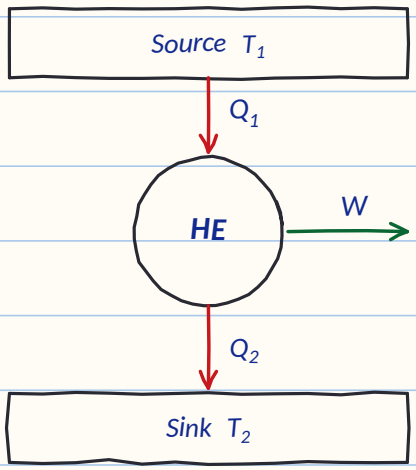
→ Does not tell whether a process is possible: all processes satisfying energy balance are allowed, yet many do not occur.

→ Does not distinguish between quality of energy: work and heat treated equal although heat cannot be completely converted to work.

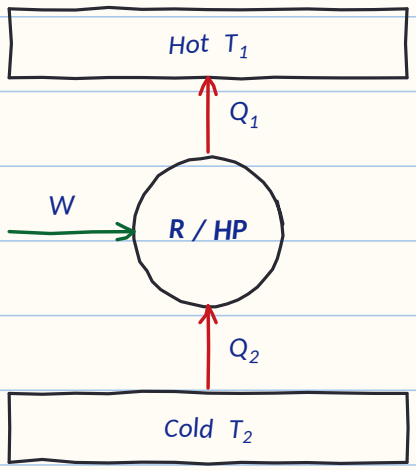
→ Does not give the limit of conversion of heat to work (efficiency) – needs the Second Law.

12. Thermal Reservoir, Heat Engine, Heat Pump

- Thermal energy reservoir (TER): very large body whose temperature does not change when heat is added or removed. Source supplies heat; sink absorbs heat.
- Heat engine: cyclic device that receives heat Q_1 from a high-T source, converts part to work W, and rejects Q_2 to a low-T sink.
- Refrigerator: device operating in cycle that extracts Q_2 from a cold space and rejects Q_1 to a warmer surrounding using work input. Heat pump: same cycle, but purpose is to supply Q_1 to the hot space.



Heat engine ($T_1 > T_2$)



Refrigerator / Heat pump

Parameters of performance

$$\eta_{th} = W_{net}/Q_1 = 1 - Q_2/Q_1$$

$$COP_R = Q_2/W = Q_2/(Q_1 - Q_2)$$

$$COP_{HP} = Q_1/W = Q_1/(Q_1 - Q_2)$$

$$COP_{HP} = COP_R + 1$$

- $\eta < 1$ always ($Q_2 \neq 0$); COP_R can be > 1; $COP_{HP} > 1$ always.
- 1 ton of refrigeration = 3.517 kW = 211 kJ/min (heat to freeze 1 ton of water at 0 °C in 24 h).
- Other measures: EER = capacity (Btu/h) / power (W); specific fuel consumption for engines.

13. Solved Numerical Problems

Problem 1

A system receives 100 kJ of heat and does 40 kJ of work. Find ΔU.

$$\Delta U = Q - W = 100 - 40 = 60 \text{ kJ}$$

Problem 2 : cycle of three processes

1-2: $Q = 40$, $W = 15$ kJ; 2-3: $Q = -35$, $\Delta U = -10$ kJ; 3-1: $W = -10$ kJ. Find the unknowns.

$$\Delta U_{12} = 40 - 15 = 25\ kJ\ ;\ W_{23} = Q - \Delta U = -35 + 10 = -25\ kJ$$

$$\Delta U_{31} = -(25 - 10) = -15\ kJ\ ;\ Q_{31} = -15 + (-10) = -25\ kJ$$

$$\Sigma Q = 40 - 35 - 25 = -20\ kJ = \Sigma W = 15 - 25 - 10 = -20\ kJ\ \checkmark$$

Problem 3 : constant pressure heating of air

1 kg air heated from 300 K to 400 K at constant pressure. Find Q, W, ΔU.

$$Q = c_p\Delta T = 1.005\times100 = 100.5\ kJ\ ;\ W = R\Delta T = 28.7\ kJ$$

$$\Delta U = c_v\Delta T = 0.718\times100 = 71.8\ kJ\ \ (= Q - W)$$

Problem 4 : adiabatic compression

1 kg air at 1 bar, 300 K is compressed adiabatically to 6 bar. Find T_2 and work.

$$T_2 = 300 \times 6^{0.2857} = 500.6\ K$$

$$W = c_v(T_1 - T_2) = 0.718 \times (-200.6) = -144.0\ kJ$$

Problem 5 : nozzle

Steam enters a nozzle with negligible velocity at h_1 = 3000 kJ/kg and leaves at h_2 = 2800 kJ/kg. Find exit velocity.

$$V_2 = \sqrt{[2(h_1 - h_2)]} = \sqrt{(2 \times 200 \times 1000)} = 632.5\ m/s$$

Problem 6 : steam turbine

$\dot{m}$ = 5 kg/s, h_1 = 3200, h_2 = 2400 kJ/kg, heat loss 40 kW. Find power output.

$$\dot{W} = \dot{m}(h_1 - h_2) - \dot{Q}_{loss} = 5 \times 800 - 40 = 3960\ kW$$

Problem 7 : air compressor

$\dot{m}$ = 0.5 kg/s, h_1 = 300, h_2 = 450 kJ/kg, heat loss 10 kW. Find power input.

$$\dot{W} = \dot{Q} - \dot{m}(h_2 - h_1) = -10 - 0.5 \times 150 = -85\ kW\ \ (85\ kW\ input)$$

Problem 8 : engine, refrigerator, heat pump

Engine: $Q_1 = 1000$, $W = 400$ kJ $\Rightarrow \eta = 40\,\%$, $Q_2 = 600$ kJ

Refrigerator: 10 kW cooling, COP = 4 $\Rightarrow$ $W = 2.5$ kW, $Q_1 = 12.5$ kW

Heat pump: $COP_{HP} = COP_R + 1 = 5$

14. Derivations to Remember

(a) SFEE derivation – steps

→ Take a control volume with inlet 1 and outlet 2, steady flow. Energy entering = energy leaving.

→ Energy carried by 1 kg of flowing fluid = $u + V^2/2 + gz$ plus flow work $P\,v$ needed to push it in.

$$\dot{m}(u_1 + P_1v_1 + V_1^2/2 + gz_1) + \dot{Q} = \dot{m}(u_2 + P_2v_2 + V_2^2/2 + gz_2) + \dot{W}_s$$

→ Since $h = u + Pv$, we get $\dot{Q} - \dot{W}_s = \dot{m}[(h_2 - h_1) + (V_2^2 - V_1^2)/2 + g(z_2 - z_1)]$.

(b) Work in an adiabatic process

For $Q = 0$ the first law gives $W = -\Delta U = m\,c_v(T_1 - T_2)$. With $PV = mRT$ and $c_v = R/(\gamma-1)$:

$$W = m\,R\,(T_1 - T_2)/(\gamma - 1) = (P_1V_1 - P_2V_2)/(\gamma - 1)$$

(c) Work in an isothermal process

$$W = \int P\,dV = \int (mRT/V)\,dV = m\,R\,T\,\ln(V_2/V_1) = P_1V_1\,\ln(P_1/P_2)$$

(d) Relation $c_p - c_v = R$

For an ideal gas $h = u + RT \Rightarrow dh/dT = du/dT + R \Rightarrow c_p = c_v + R$.

15. Unsteady (Transient) Flow – Brief Idea

In filling or emptying a tank, the mass inside the control volume changes with time, so SFEE cannot be used. The energy balance over the process is:

$$Q - W = m_2u_2 - m_1u_1 + \Sigma\, m_eh_e - \Sigma\, m_ih_i$$

- Rigid insulated tank filled from a line at enthalpy h_i (no work, no heat): $m_2u_2 - m_1u_1 = m_ih_i$.
- For an ideal gas filling an evacuated tank: $T_{final} = \gamma\, T_{line}$. The flow work gets converted into internal energy.

16. More Solved Problems

Problem 9 : polytropic expansion of air

1 kg air expands polytropically ($n = 1.3$) from 5 bar, 400 K to 1 bar. Find W, ΔU and Q.

$$T_2 = 400\,(1/5)^{0.3/1.3} = 276\ K$$

$$W = R(T_1 - T_2)/(n-1) = 0.287 \times 124/0.3 = 118.6\ kJ$$

$$\Delta U = c_v(T_2 - T_1) = 0.718 \times (-124) = -89.1\ kJ$$

$$Q = \Delta U + W = +29.6\ kJ \quad \text{(heat is added)}$$

Problem 10 : stirring work in a rigid tank

A rigid insulated-then-leaky tank is stirred with 100 kJ of paddle work while 30 kJ of heat leaves. Find ΔU.

$$\Delta U = Q - W = (-30) - (-100) = +70\ kJ$$

Problem 11 : heat pump for heating

A house loses 30 kW; heat pump $COP_{HP} = 4$. Power required?

$$W = Q_1/COP_{HP} = 30/4 = 7.5\ kW \quad (22.5\ kW \text{ taken free from outside air})$$

Problem 12 : liquid heated in a heat exchanger

Water 2 kg/s, 20 → 60 °C ($c = 4.18$ kJ/kg·K):

$$\dot{Q} = \dot{m}\, c\, \Delta T = 2 \times 4.18 \times 40 = 334.4\ kW$$

17. Glossary & Short Answer Questions

→ Adiabatic wall – no heat crosses; diathermic wall – allows heat transfer.

→ Why is Q – W = ΔE valid for irreversible processes? It only balances energy, no assumption of reversibility.

→ Why is enthalpy useful? Combines U and flow work; Q = ΔH at constant P, and appears directly in SFEE.

→ Is the work in free expansion zero? Yes, because there is no resisting force (P_{ext} = 0).

→ Define heat pump COP. COP_{HP} = desired heating / work input = Q_1/W.

→ Is heat a property? No – a path function and energy in transit.

→ What does zeroth law establish? Temperature as a property and equality of temperature for thermal equilibrium.

18. Quick Revision & Important Questions

- Q – W = ΔE (closed); SFEE: q – w = Δh + ΔKE + ΔPE (open).
- Heat & work: path functions. U, H: properties. PMM-I impossible.
- h = u + Pv; c_p – c_v = R. Throttling is isenthalpic; nozzle increases KE.

→ State and explain the first law for a closed system undergoing a cycle and a process.

→ Describe Joule's experiment. What does it prove?

→ Derive the steady flow energy equation and apply to nozzle, turbine and compressor.

→ Define PMM-I. Why is it impossible?

→ What are the limitations of the first law? Differentiate heat engine, refrigerator and heat pump.

~ End of Unit II ~